

```

69  if (touch.available()) {
70      int touchX = touch.data.x;
71      int touchY = touch.data.y;
72
73      for (int row = 0; row < 4; row++) {
74          for (int col = 0; col < 6; col++) {
75              int keyX = (tft.width() - keyboardWidth) / 2 + col * keySizeX;
76              int keyY = (tft.height() - keyboardHeight) / 3 + row * keySizeY;
77
78              if (touchX >= keyX && touchX < keyX + keySizeX && touchY >= keyY && touchY < keyY + keySizeY) {
79                  char touchedKey = 'A' + (row * 6) + col;
80                  Serial.println("Touched key: " + String(touchedKey));
81                  typedText += touchedKey;
82                  displayTypedText();
83              }
84          }
85      }
86  }
87  }
88
89  void displayTypedText() {
90      tft.setTextSize(3);
91      int textWidth = tft.textWidth(typedText);
92      int textX = (tft.width() - textWidth) / 2;
93      int textY = (tft.height() - keyboardHeight) / 3 - keySizeY; // Above the keyboard
94      tft.setCursor(textX, textY);
95      tft.setTextColor(TFT_WHITE);
96      tft.print(typedText);
97      Keyboard.write(typedText);
98  }
99  }

```

Fig. 8: Code snippet to detect the touch point and send as HID input

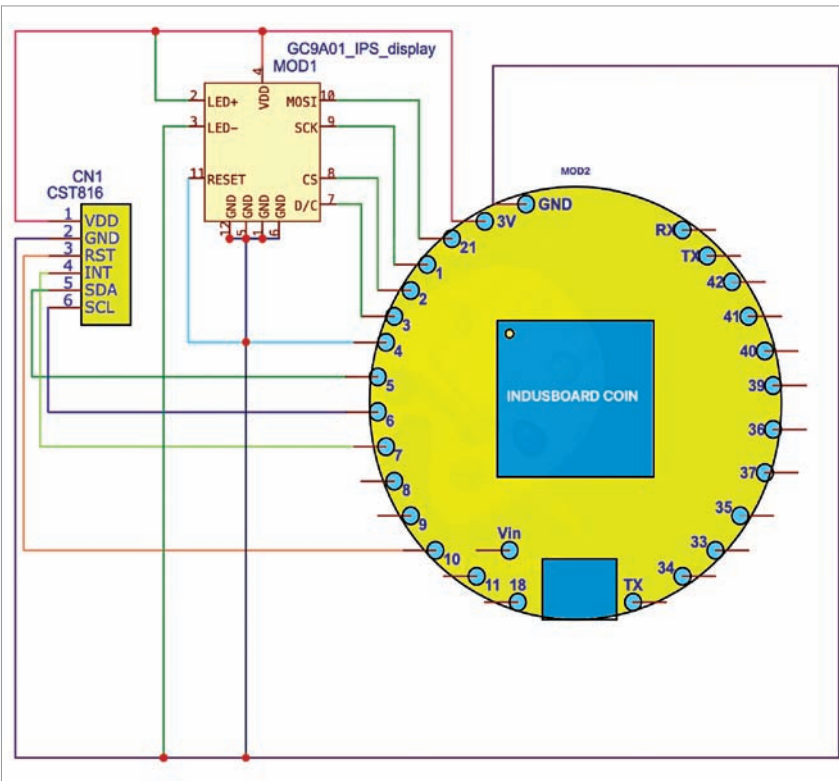


Fig. 9: Circuit diagram

Here, the pins 5 and 6 are used. Refer to Fig. 6 for code settings.

Create the keyboard layout, display it on the keyboard, map the touchpoints to the displayed keys, and detect touch on those keys. Then, send that letter as HID input

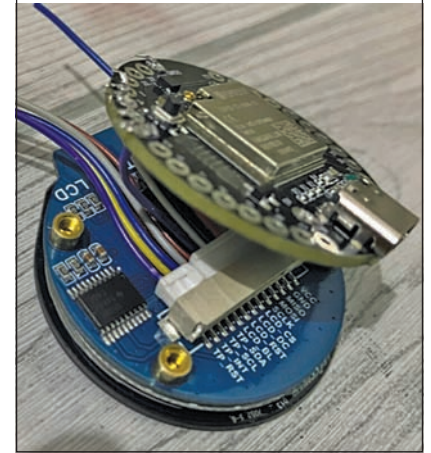
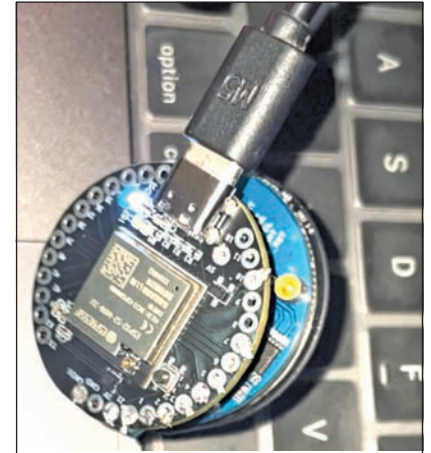


Fig. 10: Indusboard soldered with the display

EFY Note

The source code of this project is available for free download at www.electronicsforu.com